Optimal Reconfiguration of Systems in Aeronautics

Lecture 03: "System Diagnosis"

Bogdan D. Ciubotaru

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Outline

1 System Diagnosis

Objective I

This lecture introduces the basics of system diagnosis.

Outline

1. System Diagnosis
   - Nominal Operation
   - Faulty Behavior
   - Modeling of Impaired Systems
   - Fault Detection, Isolation, and Identification / Estimation

Introduction I

Automatic Control, as hidden-technology, aims at transparently ensuring that the behavior of the controlled system satisfies a given set of specifications for which it was designed under the nominal conditions.

Thus, in order to solve an ordinary control problem for a certain plant, a series of abstract entities are to be identified, i.e., a control objective O, a class of admissible control laws $\mathscr{U}$, and a set of constraints C.

The constraints are functional relations that describe the behavior of the dynamic system, which also define, in turn, a structure S and parameters θ.

General control problem I

In terms of notations introduced before, the General Control Problem (GCP) is defined as simple as in the following definition.

Concerning the fault tolerant control analysis that is to be developed, notations introduced before are identified through their index n, in nominal evolution, and through index f, in post-fault operation, respectively.

Definition (GCP - General Control Problem)

Solve

$$P^g_{c_n} \triangleq \langle O, C, \mathscr{U} \rangle \,. \tag{1}$$

□

In words, solving the general control problem (1) means to find in $\mathscr{U}$ a control law U that satisfies C while achieving O; moreover, some performance index J might be associated with a control objective O and, when several solutions exist, the best one is selected according to J, naming it the optimal one.

Standard control problem I

Moreover, the explicit form of GCP, namely the Standard Control Problem (SCP), is derived in the following definition.

Definition (SCP - Standard Control Problem)

Solve

$$P^{s}_{c_n} \triangleq \langle O, S, \theta, \mathscr{U} \rangle \,. \tag{2}$$

□

Precisely, solving the standard control problem (2) means to find in $\mathscr{U}$ a control law U that achieves O for the system with structure S and behavior characterized by the particular set of parameters θ.

However, the assumption that the system behavior is fairly described only by the static parameters θ is too restrictive, since the identification process has a certain degree of uncertainty, that is the determination of θ is assumed to have errors, or they may be time-varying, $\theta(t)$.

In that case, a more flexible view would impose that parameters θ take values inside a set Θ.

Robust control problem I

In these conditions, achieving O under constraints whose structure is S and whose parameters belong to Θ transforms SCP from (2) into a robust formulation, which stands for minimizing over Θ the discrepancy of the achieved results.

Using the same notation as for SCP, the Robust Control Problem (RCP) is described in the following definition.

Definition (RCP - Robust Control Problem)

Solve

$$P^r_{c_n} \triangleq \langle O, S, \Theta, \mathscr{U} \rangle \,, \tag{3}$$

where Θ stands for a set of possible values of θ. $\square$

Anyhow, searching for the RCP solution over the range limited by Θ is not the only plausible approach; yet another technique that might be used is the identification / estimation of the real values $\hat{\theta}$ of uncertain θ or time-varying ones $\theta(t)$, obtaining the solution to the SCP associated with $\hat{\theta}$ afterwards.

Adaptive control problem I

This series of control actions ("parameter estimation, first - solution to the standard problem, then") has an adaptive pattern.

Always in the quadruple-notation $\langle . , . , . , . \rangle$, as for RCP, the Adaptive Control Problem (ACP) is formulated in the following definition.

Definition (ACP - Adaptive Control Problem)

Solve

$$P^a_{c_n} \triangleq \langle O, S, \hat{\theta}, \mathcal{U} \rangle \,, \tag{4}$$

where the estimation of $\hat{\theta}$ is part of the adaptation. $\square$

Preliminaries I

However, due to different known or unknown factors, presumed or not, but other than the usual influences, namely perturbations and noise, the system operation is exposed to the risk that its nominal evolution steps slightly or in an abrupt manner to an undesired one; the change to this new situation is called faulty behavior.

Regarding this, constraints C_n defined by structure S_n and parameters θ_n change to C_f, respectively S_f and θ_f; moreover, the class of nominal admissible controls $\mathscr{U}_n$ might also be different than the post-fault admissible one $\mathscr{U}_f$.

Thus, unchanged is only the objective, that is $O_n \equiv O_f$, namely the control system must be fault tolerant, meaning to achieve the same control objective in both operations, respectively nominal and faulty.

System fault I

In what follows, the system fault, notion which identifies the primary concern of the Fault Detection, Identification and Isolation / Estimation (FDIE) research, is defined as informal as in the following definition.

Definition (System Fault)

The system fault is defined as an unpermitted deviation of at least one characteristic property of a system variable from an acceptable behavior. □

Therefore, the fault may lead to a malfunction or failure of the system; however, distinction should be made between the notions of malfunction, which assumes acceptable performance degradation of the system but not catastrophic behavior, and that of failure, which defines the system state from which continuing operation is impossible. (At any rate, the nomenclature used and the definitions introduced are subjective to the interpretation of different research groups.)

Of course, the informal description of the system fault from the definition is only of general use; hence, a mathematical approach to specifying the different types of faults which are possible to occur in the classical parts of an automatic control system, precisely sensors, actuators, or the plant itself, is to be detailed in the sequel. (It is assumed that other types of impairments, like software codes, electrical cables, or transmission networks in the automatic system, are transparent or reflected in the classical components.)

Preliminaries I

It is a known fact that, using the physical laws, most of the engineering systems can be modeled by the system of differential equations

$$\begin{aligned} \dot{x}(t) &= g\left[x(t), u(t), d(t), f(t)\right] , \\ y(t) &= h\left[x(t), u(t), d(t), f(t)\right] , \end{aligned} \tag{5}$$

where g and h are nonlinear functions, $d(t)$ is the vector of unknown inputs indicating the vector of disturbances, while $f(t)$ stands for the vector of faults.

Since it is quite difficult to handle the system exactly in the general nonlinear form (5), when the system is operating in nominal conditions that are characterized by an equilibrium state, its linear correspondent is obtained by performing a Taylor series expansion of functions g and h around the equilibrium point and keeping only the first order terms of these series.

Additive faults I

Hence, system (5) can be described by the linear time-invariant (LTI) model

$$\dot{x}(t) = A_n x(t) + B_n u(t) + E_d d(t) + E_f f(t)\,, \tag{6a}$$

$$y(t) = C_n x(t) + G_d d(t) + G_f f(t)\,. \tag{6b}$$

In this representation, the fault $f(t)$ appears as an additional input in the model and therefore it is called an additive fault.

In fault-free operation, $f(t)$ is zero for all t, i.e., $f(t) \equiv 0\,,\ \forall\ t \geq 0$.

However, the assumption that all the influences onto the system can be treated as additive seems being too simplistic. Since the linear system which is to be controlled represents the equilibrium around a given point of a nonlinear one, the designer should be aware of any fault that could make this linearization useless.

Multiplicative faults I

As opposed to the additive faults, multiplicative faults consist in changes in the entries of matrices A_n, B_n, and C_n.

Thus, system (5) can be described by the LTI model

$$\dot{x}(t) = A_f x(t) + B_f u(t) + E_d d(t)\,, \tag{7a}$$

$$y(t) = C_f x(t) + G_d d(t)\,. \tag{7b}$$

Naturally, a common way to test whether there is something going wrong in the system is to take the system nominal model and compare it with that actually in operation, in terms of state trajectory for the state-space representation. Anyhow, to avoid hazardous evolutions of the control system, the continuous monitoring of measurable variables and the verification of their domains of variation limits appear to be simple but is not proficient for complex systems.

In this regard, a technical diagnosis system must be designed and implemented such that to detect all the unusual changes in the system operation and to make their characterization as in (6) respectively (7), either suggesting possible counteractions to human operators or starting and managing automatic remedial procedures.

FDIE modules I

Defining the basic features of such a diagnosis system means to distinguish between the modules composing it according to their functionalities, namely Fault Detection, Fault Isolation, and Fault Identification / Estimation.

These are described as follows:

- Fault Detection: it decides whether or not a fault has occurred; it is represented by a binary decision in terms of "yes / no"-type answer.

- Fault Isolation: it finds in which component / particular subsystem, namely sensor, actuator, or plant, the impairment produced.

- Fault Identification / Estimation: it identifies the fault model and estimates the numerical values of its parameters; also, it determines the time-instant at which the fault model becomes available.

Residual signal I

From the previous characterization, the residual is identified as the second important concern of the FDIE research, notion which is to be introduced in the following definition.

Definition (Residual)

The residual is defined as a linear or nonlinear function of the inputs and outputs of the monitored system; explicitly, residuals $r(t)$ are variables that define the degree of consistency between the behavior of the real plant and that of the model; in the fault-free case, the residual is zero or near to zero, very small in some sense; when a fault occurs in the system, the residual grows significantly. □

In the sequel, the modules of the FDIE system are characterized in detail.

Fault detection I

The first step of the FDIE investigation is fault detection, which, in turn, consists of another two stages, namely Residual-Generation and Decision-Making (Residual-Evaluation).

These are described as follows:

- Residual-Generation: it uses the model and the input-/output- recordings to determine the residuals $r(t)$.

- Residual-Evaluation: it works on the residuals $r(t)$, which are evaluated in order to effectively detect, isolate, and identify faults $f(t)$ (naturally, in both steps, model uncertainties, disturbances, and measurement noise have to be taken into account).

Fault detection II

Obviously, if there is no fault in the system, $f(t) = 0$, and the residual $r(t)$ is robust against disturbances $d(t)$, the basic condition of residual-generation assumes

$$r(t) \neq 0 \;\Rightarrow\; f(t) \neq 0\,. \tag{8}$$

Consequently, the fault $f(t)$ is effectively detected after comparing some residual-evaluation function $J(r(t))$ with a threshold function $T(t)$ according to the test

$$\begin{aligned} J[r(t)] &\leq T(t)\,, \;\text{ for } f(t) = 0\,, \\ J[r(t)] &> T(t)\,, \;\text{ for } f(t) \neq 0\,; \end{aligned} \tag{9}$$

regarding this test, a fault is likely if the threshold is exceeded by the residual-evaluation function.

Fault detection III

At the design stage, a positive constant can be used for $T(t)$ and $J[r(t)]$ may be chosen as the residual-norm vector; however, if this limit is chosen too high, then the number of missed-faults will be considerable, as it would be the number of false-alarms, if the limit is set too low. An adaptive threshold is the most indicated, a possible choice being in calculating it by integration of the residual variations over a time window.

It should be noted that, for different types of system modeling, particular parameterizations of the residual generator are possible. Anyhow, the necessary and sufficient conditions for the existence of such signal processor are the classical rank-observability conditions, which state that the maximum number of disturbances that can be decoupled cannot be larger than the number of independent measurements.

Fault isolation I

The successful fault detection is usually followed by fault isolation, operation which must distinguish that particular fault from others. Whether a single residual signal is sufficient to detect faults, a set or vector of residuals is required for fault isolation.

In this regard, there exist two different ways in constructing the set of residuals in order to achieve fault isolation, namely Structured- and Directional- Residuals.

These are described as follows:

- Structured-Residuals Set: it considers that each residual is designed to be sensitive to a subset of faults, while remaining insensitive to other faults; the design procedure consists of two steps: the first is to specify the sensitive-insensitive relationships between residuals and faults according to the isolation task and the second is to design a set of residual-generators according to the desired sensitive-insensitive relationships; the fault isolation is achieved after determining which residual is nonzero.

- Directional-Residuals Set: it assumes that the residual-vector is being designed in a fixed fault-specified direction or subspace in the residual-space in response to a particular fault; the fault isolation is accomplished when determining which of the known fault-signature directions is the closest in terms of distance-norms to the generated residual-vector.

Fault identification / estimation I

The fault models used to redesign the new control law or to adapt the existing one may be identified offline based on the Failure Modes and Effects Analysis (FMEA), or online by using algorithms of Recursive Least Squares (RLS) or Fast Fourier Transforms (FFT).

In what follows, the state-space parameterizations of sensor and actuator faults are presented.

Models for sensor faults I

Regarding (7b), the sensors nominal operation, namely the no-fault case, is indicated by the fact that the measurements are identical with the values of the states, that is

$$y_{f_i}(t) = x_i(t)\,,\ \forall\ t \geq 0\,, \tag{10}$$

assuming that $C_f = I_n$, while the common sensor faults can be classified in the following categories (in the sequel, the major focus is on the influence of faults onto the system outputs, such that the perturbations vector $d(t)$ is ignored):

Models for sensor faults II

- Bias:

$$y_{f_i}(t) = x_i(t) + f_i(t)\,,\ \dot{f}_i(t) \equiv 0\,,\ f_i(t) \neq 0\,,\ \forall\ t \geq t_{f_i}\,. \tag{11a}$$

- Drift:

$$y_{f_i}(t) = x_i(t) + f_i(t)\,,\ |f_i(t)| = l_i t\,,\ 0 < l_i \ll 1\,,\ \forall\ t \geq t_{f_i}\,. \tag{11b}$$

- Loss-of-Accuracy (Performance-Degradation):

$$y_{f_i}(t) = x_i(t) + f_i(t)\,,\ |f_i(t)| \leq \bar{f}_i\,,\ \dot{f}_i(t) \to 0\,,\ \forall\ t \geq t_{f_i}\,. \tag{11c}$$

- Freezing:

$$y_{f_i}(t) = x_i(t_{f_i})\,,\ \forall\ t \geq t_{f_i}\,. \tag{11d}$$

- Loss-of-Effectiveness (Calibration-Error):

$$y_{f_i}(t) = c_{f_i} x_i(t)\,,\ \forall t \geq t_{f_i}\,. \tag{11e}$$

Models for sensor faults III

Above, t_{f_i} denotes the time-instant of the i-th sensor fault, f_i denotes its accuracy-coefficient such that $0 < f_i \in [f_i^{\min}, f_i^{\max}]$, and the effectiveness-coefficient satisfies $c_{f_i} \in (0, 1)$; usually, the sensor faults modeled as in (11e) are reflected in the elements of matrix C_f from (7b).

The effect of sensor faults is that they break or corrupt the information link between the plant and the controller, which may drive the plant partially unobservable.

Models for actuator faults I

Moreover, regarding (7a), the actuators nominal operation, namely the no-fault case, is indicated by the fact that the post-fault control law is identical with the unimpaired one, that is

$$u_{f_i}(t) = u_{n_i}(t)\,,\ \forall\ t \geq 0\,, \tag{12}$$

while the typical control effector faults can be classified as above in the following categories:

Models for actuator faults II

- Lock-in-Place:

$$u_{f_i}(t) = u_{n_i}(t_{f_i}) \,, \;\; \forall \; t \geq t_{f_i} \,; \tag{13a}$$

the effector freezes at a certain condition and does not respond to subsequent commands.

- Hard-Over:

$$u_{f_i}(t) = u_{n_i}^{\min} \text{ or } u_{f_i}(t) = u_{n_i}^{\max} \,, \;\; \forall \; t \geq t_{f_i} \,; \tag{13b}$$

the effector is moving to the upper- or lower- limit position regardless of the command; the speed of response is limited by the effector rate limit.

- Float:

$$u_{f_i}(t) = 0 \,, \;\; \forall \; t \geq t_{f_i} \,; \tag{13c}$$

the effector floats with zero-moment and does not contribute to the control authority.

- Loss-of-Effectiveness:

$$u_{f_i}(t) = (1 - \tau_{f_i}) u_{n_i}(t) \,, \;\; \forall \; t \geq t_{f_i} \,; \tag{13d}$$

the effector gain is lowering w.r.t. its nominal value.

Models for actuator faults III

Above, u_n represents the nominal command computed by the controller, t_{f_i} denotes the time-instant of the i-th effector fault and its effectiveness-coefficient τ_{f_i} satisfies $\tau_{f_i} \in (0, 1)$; usually, the actuator faults as in (13d) are reflected in the elements of matrix B_f from (7a).

The effect of actuator faults is that they affect the way in which the plant is conducted through the control law, which can make the plant partially uncontrollable.

Models for structural / process faults I

The structural faults are seen as changes in the dynamical behavior of the process but they appear to be more difficult to parameterize; in what follows, it will be assumed that an identification / estimation procedure of the post-fault situation is able to provide an expression for matrix A_f from (7a).

In these lectures, sensor faults are not taken into consideration but the main concern is represented by the Loss-of-Effectiveness actuator fault that changes the control effort distribution matrix from B_n to

$$B_f = (1 - \tau_f) B_n \tag{14}$$

and by the structural / process faults that change the system matrix from A_n to

$$A_f = (1 - \gamma_f) A_n \text{ or} \tag{15a}$$

$$A_f = A_n + \Delta A_f . \tag{15b}$$

Questions? vs. Answers!

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